

MCSA-P02

Project Evidence

Controlled Document Centre

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Lab Environment: VMware Virtual Environment

Operating System: Microsoft Windows Server 2016

Domain: Aster.com

Technologies Implemented

- Active Directory Domain Services (AD DS)
- DNS
- DHCP
- SMB File Sharing
- NTFS Permissions
- Group Policy
- Windows Server Backup
- Windows Firewall
- Event Logs

Project Focus

Centralized identity management • Network services • File security • Group Policy • Backup & Recovery

Project Type: Hands-on Training / Lab Project

Server Infrastructure: AST-DC01, AST-FS01, AST-BK01, AST-CL01, AST-FS02

Project Overview

The Controlled Document Centre is a hands-on Windows Server lab project focused on implementing centralized and controlled document storage in a domain-based environment. The project demonstrates how organizations can use Active Directory, security groups, NTFS permissions, and SMB file sharing to manage access to departmental documents securely.

Business Scenario

The simulated organization required a centralized infrastructure to:

- Provide named user accounts through Active Directory.
- Control access to departmental files.
- Provide network configuration through DHCP.
- Provide DNS name resolution.
- Apply organization-wide security policies.
- Protect important files using backup and recovery.
- Maintain appropriate access to Finance, HR and Operations resources.

Infrastructure Components

Server	Role	IP Address
AST-DC01	Domain Controller, AD DS, DNS, DHCP	192.168.50.10
AST-FS01	File Server / SMB Shares	192.168.50.20
AST-BK01	Backup Server	192.168.50.30
AST-CL01	Client / Validation Machine	192.168.50.90
AST-FS02	File Server	192.168.50.21

Main Implementation Areas

1. Active Directory

Created the Aster.com domain structure, organizational units, users and security groups.

2. DNS

Configured the Aster.com DNS zone and server records to support name resolution.

3. DHCP

Configured a DHCP scope with the address range **192.168.50.100–192.168.50.149**.

4. File Services

Created departmental folders for **Finance, HR and Operations** and configured SMB shares.

5. NTFS Permissions

Applied role-based permissions using security groups to control access to departmental data.

6. Group Policy

Configured password and account-lockout policies to improve account security.

7. Backup & Recovery

Configured Windows Server Backup and validated successful backup and file-recovery operations.

Project Outcome

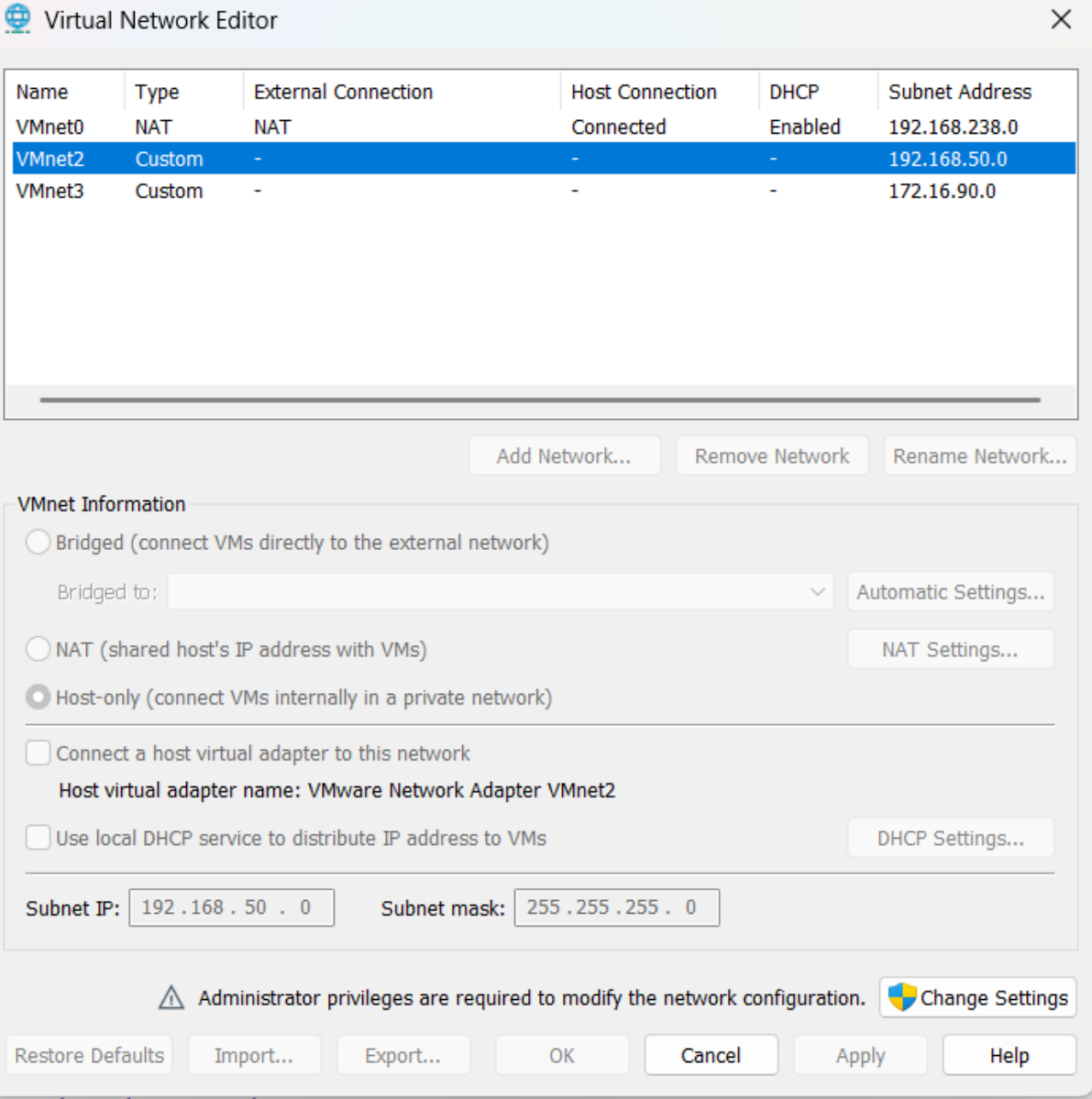
The completed lab demonstrates an integrated Windows Server environment where **identity, DNS, DHCP, file access, security policies and backup/recovery** work together as core IT infrastructure components.

Skills Demonstrated

- Windows Server Administration
- Active Directory Administration
- DNS & DHCP Configuration
- User and Group Management
- SMB File Sharing
- NTFS Permission Management
- Group Policy Configuration
- Backup & File Recovery
- Basic Troubleshooting
- VMware Virtual Lab Administration

B01 | Prepare the isolated virtual network

.bit



The screenshot shows the 'Virtual Network Editor' window. At the top, there is a table listing virtual networks. VMnet2 is selected, highlighted in blue. Below the table are buttons for 'Add Network...', 'Remove Network', and 'Rename Network...'. The 'VMnet Information' section shows three radio button options: 'Bridged', 'NAT', and 'Host-only'. 'Host-only' is selected. Below these are checkboxes for 'Connect a host virtual adapter to this network' and 'Use local DHCP service to distribute IP address to VMs'. At the bottom, there are input fields for 'Subnet IP' and 'Subnet mask'. A warning message at the bottom states: 'Administrator privileges are required to modify the network configuration.' with a 'Change Settings' button. At the very bottom of the window, there are buttons for 'Restore Defaults', 'Import...', 'Export...', 'OK', 'Cancel', 'Apply', and 'Help'.

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	NAT	NAT	Connected	Enabled	192.168.238.0
VMnet2	Custom	-	-	-	192.168.50.0
VMnet3	Custom	-	-	-	172.16.90.0

Buttons: Add Network..., Remove Network, Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)
Bridged to: Automatic Settings...

☐ NAT (shared host's IP address with VMs) NAT Settings...

☒ Host-only (connect VMs internally in a private network)

☐ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet2

☐ Use local DHCP service to distribute IP address to VMs DHCP Settings...

Subnet IP: Subnet mask:

⚠ Administrator privileges are required to modify the network configuration. [Change Settings](#)

Buttons: Restore Defaults, Import..., Export..., OK, Cancel, Apply, Help

▼ Virtual Machine Details
State: Powered off

B02 | Build AST-DC01

Server Manager

Server Manager ▸ Local Server

Dashboard

Local Server

All Servers

AD DS

DHCP

DNS

File and Storage Services ▸

PROPERTIES

For AST-DC01

Computer name	AST-DC01	Last installed updates	Never
Domain	Aster.com	Windows Update	Download updates only, using Windows Update
		Last checked for updates	Never
Windows Firewall	Public: On	Windows Defender	Real-Time Protection: On
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Ethernet0	192.168.50.10, IPv6 enabled	Product ID	00377-10000-00000-AA360 (activated)
Operating system version	Microsoft Windows Server 2016 Datacenter Evaluation	Processors	11th Gen Intel(R) Core(TM) i5-11320H @ 3.20GHz, 11th Gen Intel(R) Core(TM)
Hardware information	VMware, Inc. VMware7,1	Installed memory (RAM)	2 GB
		Total disk space	59.45 GB

EVENTS

All events | 25 total

Filter

Server Name	ID	Severity	Source	Log	Date and Time
AST-DC01	14553	Warning	Microsoft-Windows-DfsSvc	System	9/22/2026 11:11:04 AM

11:12 AM
9/22/2026

B03 | Build AST-FS01

The screenshot displays the Windows Server Manager interface for a local server named AST-FS01. The left-hand navigation pane includes links to the Dashboard, Local Server (which is currently selected), All Servers, and File and Storage Services. The main area is titled 'PROPERTIES For AST-FS01' and contains a scrollable list of system settings. These settings are organized into two columns. The first column includes Computer name (AST-FS01), Domain (Aster.com), Windows Firewall (Public: On), Remote management (Enabled), Remote Desktop (Disabled), NIC Teaming (Disabled), Ethernet0 (192.168.50.20, IPv6 enabled), Operating system version (Microsoft Windows Server 2016 Standard Evaluation), and Hardware information (VMware, Inc. VMware7,1). The second column includes Last installed updates (Never), Windows Update (Download updates only, using Windows Update), Last checked for updates (Never), Windows Defender (Real-Time Protection: On), Feedback & Diagnostics (Settings), IE Enhanced Security Configuration (On), Time zone ((UTC+05:30) Sri Jayawardenepura), Product ID (00378-00000-00000-AA739 (activated)), Processors (11th Gen Intel(R) Core(TM) i5-11320H @ 3.20GHz, 11th Gen Intel(R) Core(TM) i), Installed memory (RAM) (2 GB), and Total disk space (79.32 GB). Below the properties section is an 'EVENTS' section showing 'All events | 23 total'. It features a search filter box and a table with columns for Server Name, ID, Severity, Source, Log, and Date and Time. The first row of the table shows an event for AST-FS01 with ID 8016, a Warning severity, and a source of Microsoft-Windows-DNS-Client-Events. The Windows taskbar at the bottom shows the Start button, search icon, and several pinned applications, with the system clock indicating 11:13 AM on 9/22/2026.

Server Manager

Server Manager ▸ Local Server

Dashboard

Local Server

All Servers

File and Storage Services ▸

PROPERTIES
For AST-FS01

TASKS ▾

Computer name	AST-FS01	Last installed updates	Never
Domain	Aster.com	Windows Update	Download updates only, using Windows Update
		Last checked for updates	Never
Windows Firewall	Public: On	Windows Defender	Real-Time Protection: On
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC+05:30) Sri Jayawardenepura
Ethernet0	192.168.50.20, IPv6 enabled	Product ID	00378-00000-00000-AA739 (activated)
Operating system version	Microsoft Windows Server 2016 Standard Evaluation	Processors	11th Gen Intel(R) Core(TM) i5-11320H @ 3.20GHz, 11th Gen Intel(R) Core(TM) i
Hardware information	VMware, Inc. VMware7,1	Installed memory (RAM)	2 GB
		Total disk space	79.32 GB

EVENTS
All events | 23 total

TASKS ▾

Filter

Server Name	ID	Severity	Source	Log	Date and Time
AST-FS01	8016	Warning	Microsoft-Windows-DNS-Client-Events	System	9/22/2026 11:10:43 AM

11:13 AM
9/22/2026

B04 | Build AST-BK01

Server Manager

Server Manager ▸ Local Server

Dashboard

Local Server

All Servers

File and Storage Services ▸

PROPERTIES

For AST-BK01

Computer name	AST-BK01	Last installed updates	Never
Domain	Aster.com	Windows Update	Download u
		Last checked for updates	Never
Windows Firewall	Domain: On	Windows Defender	Real-Time P
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC-08:00)
Ethernet0	192.168.50.30, IPv6 enabled	Product ID	Not activate
Operating system version	Microsoft Windows Server 2016 Standard Evaluation	Processors	11th Gen Int
Hardware information	VMware, Inc. VMware7,1	Installed memory (RAM)	2 GB
		Total disk space	119.32 GB

EVENTS

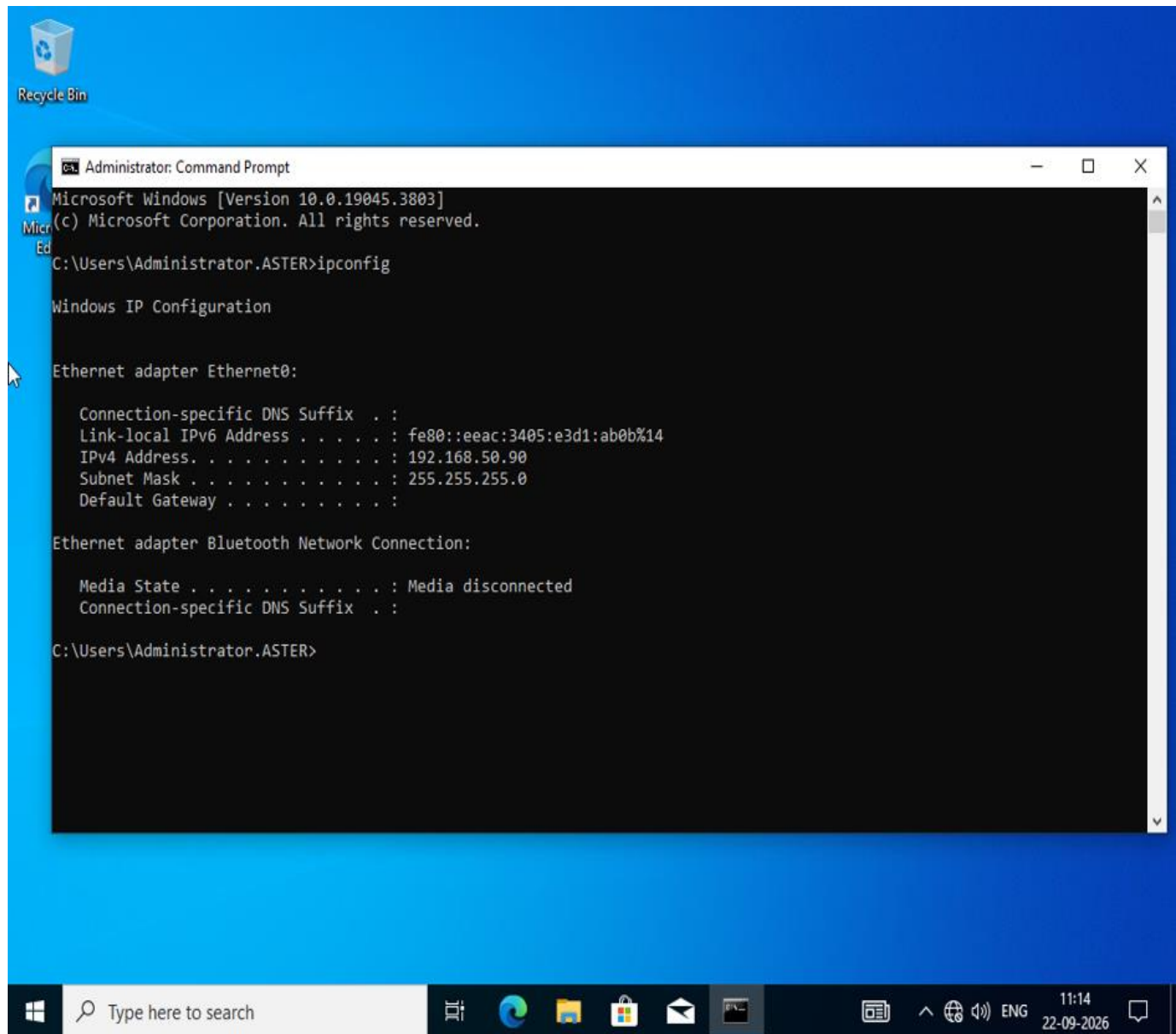
All events | 16 total

Filter

Server Name	ID	Severity	Source	Log	Date and Time
AST-BK01	8198	Error	Microsoft-Windows-Security-SPP	Application	9/21/2026 10:41:34 PM
AST-BK01	1014	Error	Microsoft-Windows-Security-SPP	Application	9/21/2026 10:41:34 PM
AST-BK01	8200	Error	Microsoft-Windows-Security-SPP	Application	9/21/2026 10:41:34 PM

10:43 PM
9/21/2026

B05 | Build AST-CL01

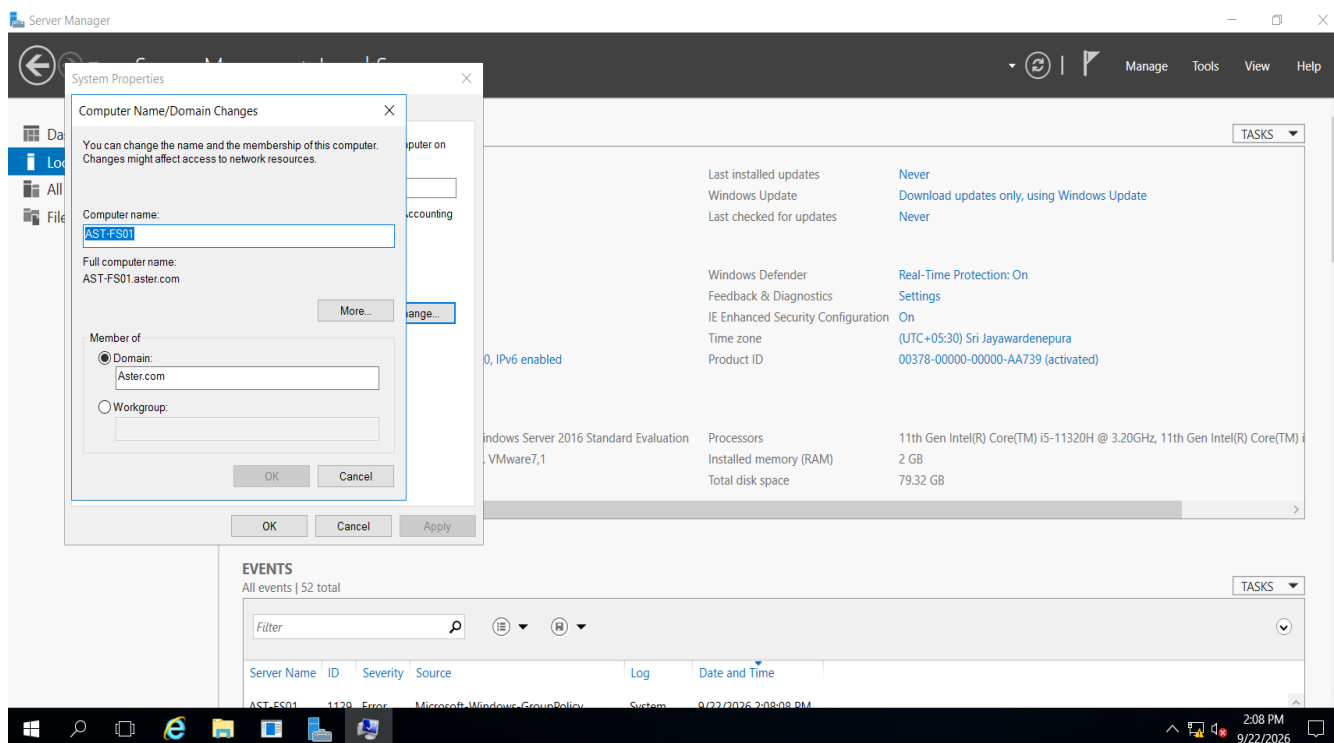


B07 | Create the forest and DNS service

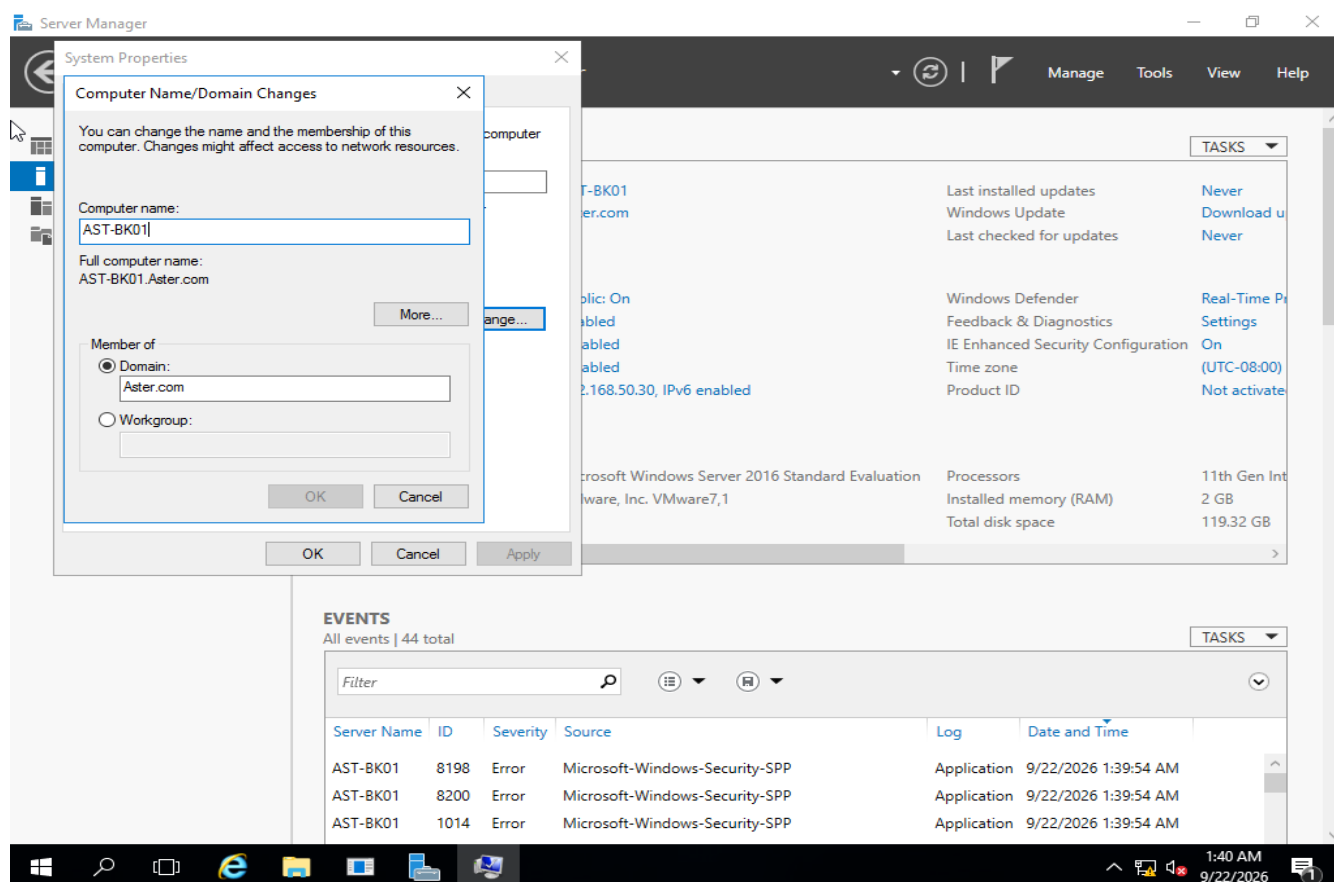
The screenshot displays the Windows DNS Manager application. The left-hand pane shows the hierarchy: DNS > AST-DC01 > Forward Lookup Zones > Aster.com. The right-hand pane shows a list of DNS records for the Aster.com zone.

Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[54] ast-dc01.aster.com, ...	static
(same as parent folder)	Name Server (NS)	ast-dc01.aster.com.	static
(same as parent folder)	Host (A)	192.168.50.10	9/18/2026 11:30:00 AM
ast-bk01	Host (A)	192.168.50.30	static
ast-cl01	Host (A)	192.168.50.90	static
ast-dc01	Host (A)	192.168.50.10	static
ast-fs01	Host (A)	192.168.50.20	static
ast-fs02	Host (A)	192.168.50.21	static
AST-VPN01	Host (A)	192.168.50.40	static

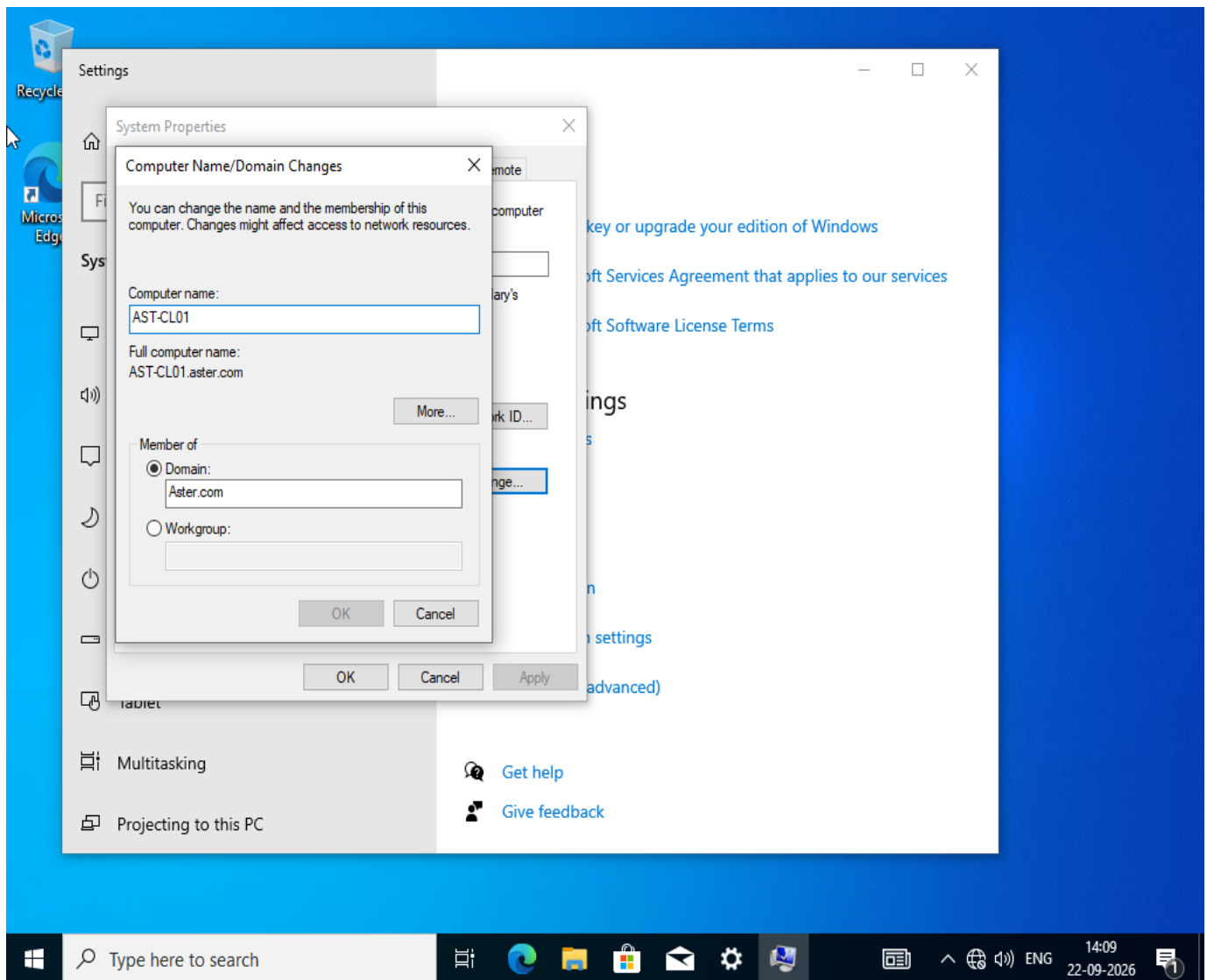
B08 | Join AST-FS01 to the domain



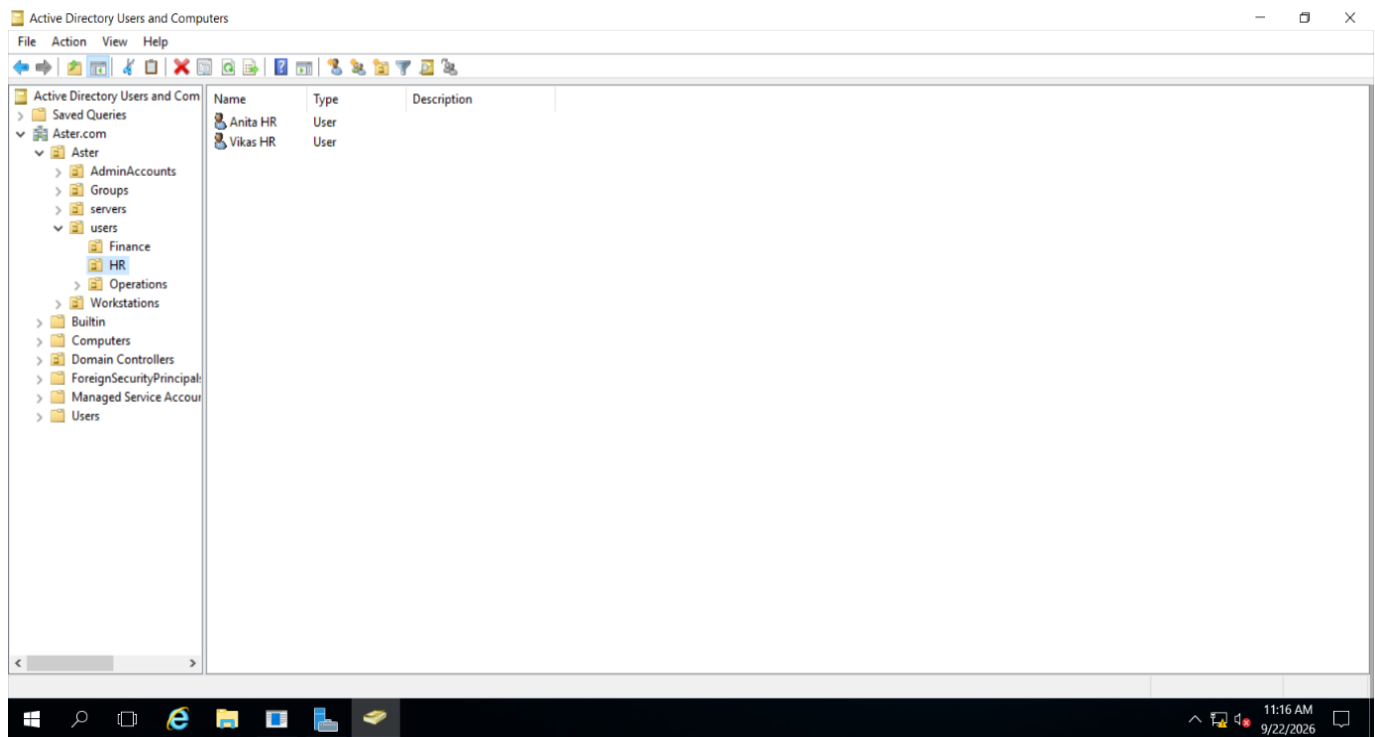
B09 | Join AST-BK01 to the domain



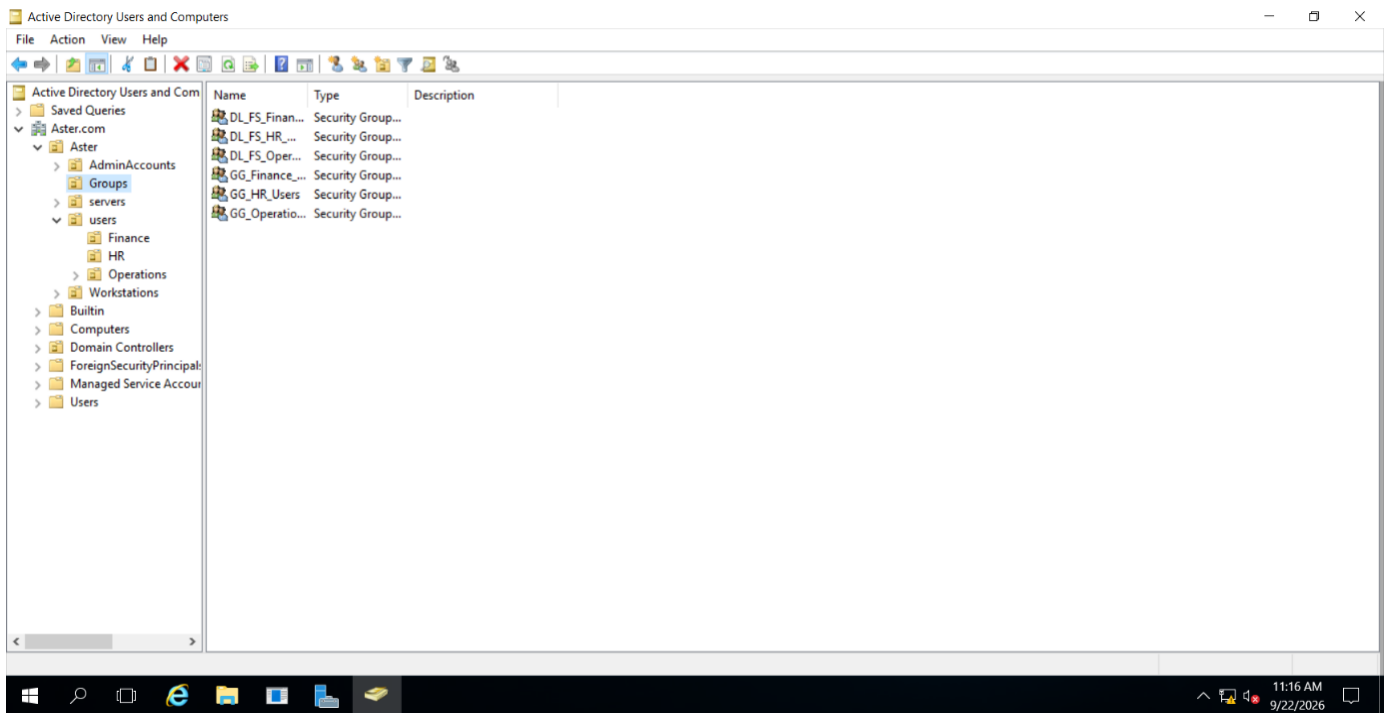
B10 | Join AST-CL01 to the domain



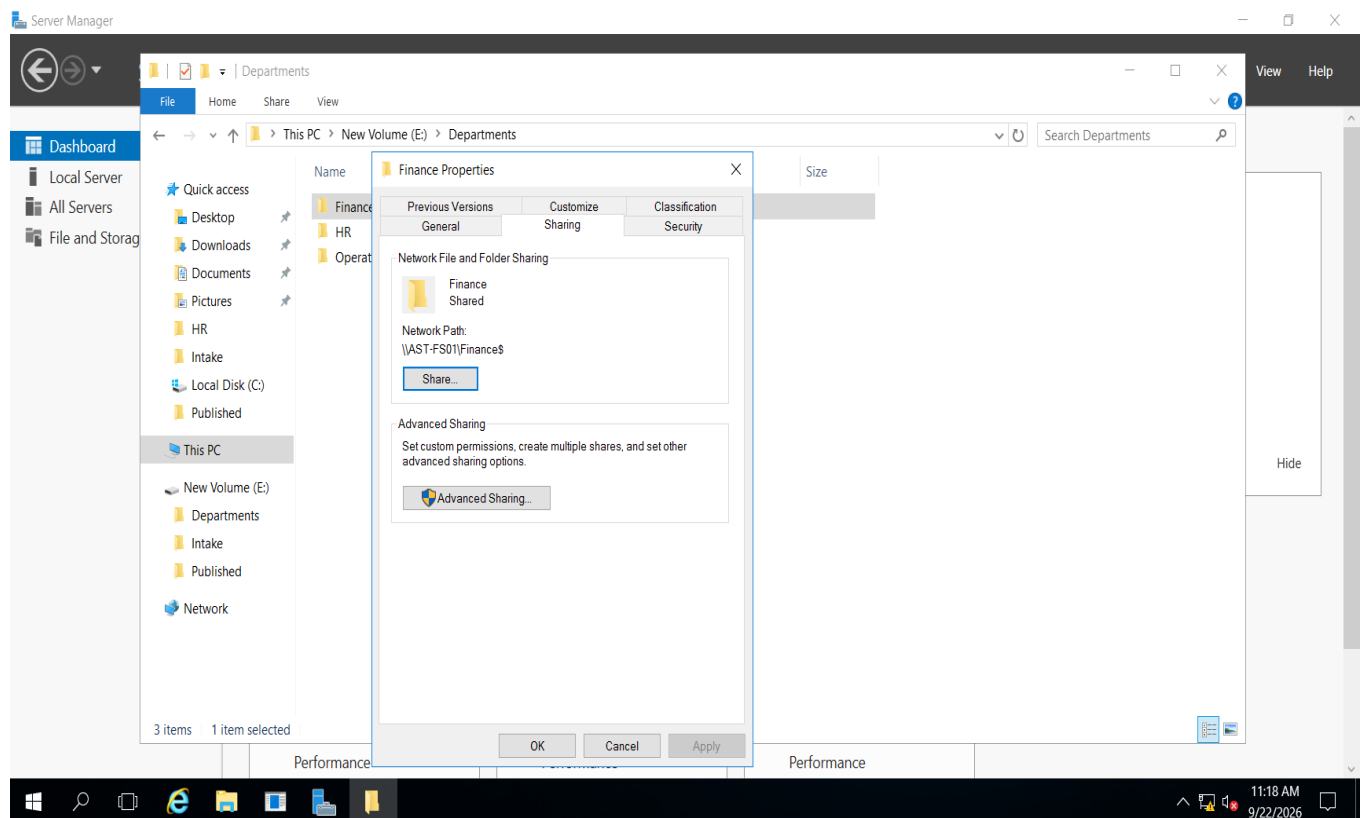
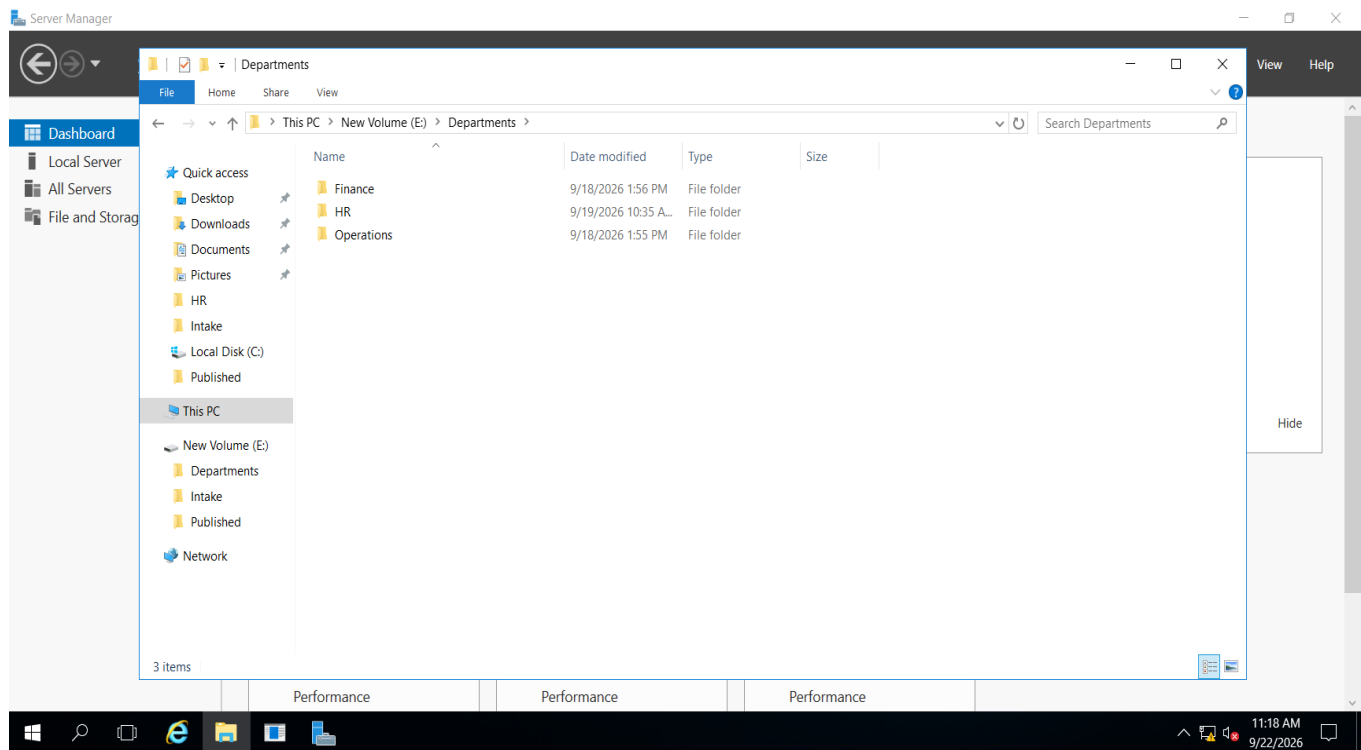
B11 | Create the organisation and named accounts

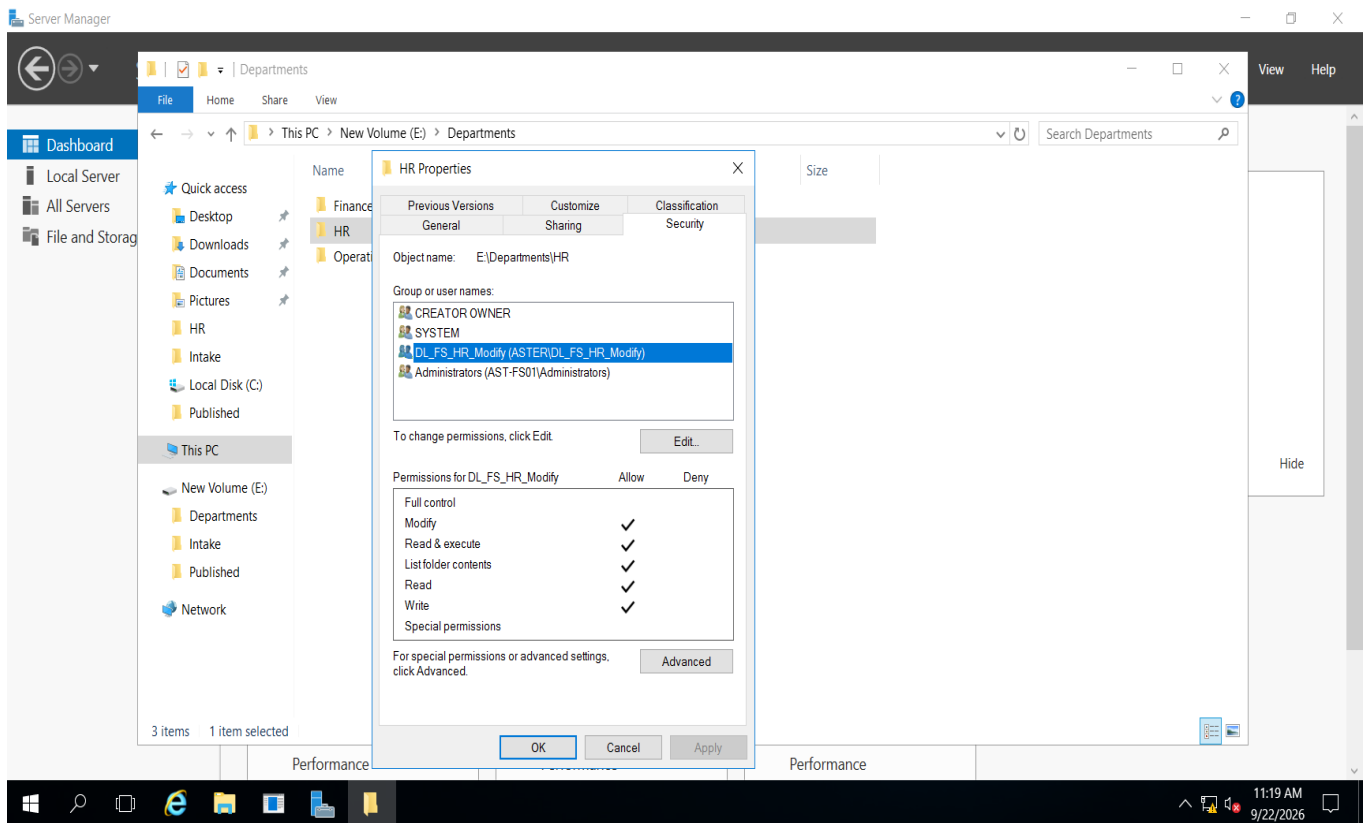


B12 | Build group based access assignments

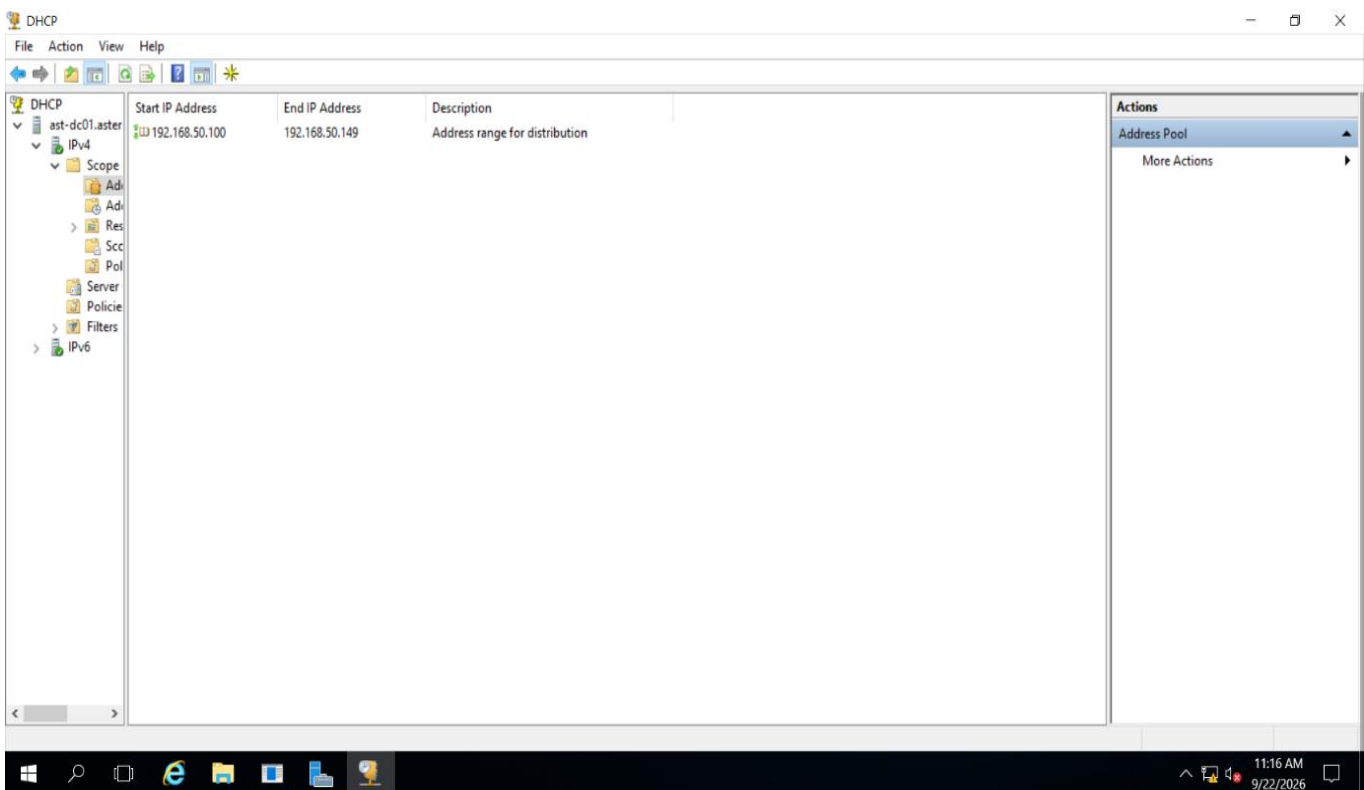


B13 | Publish three controlled departmental shares

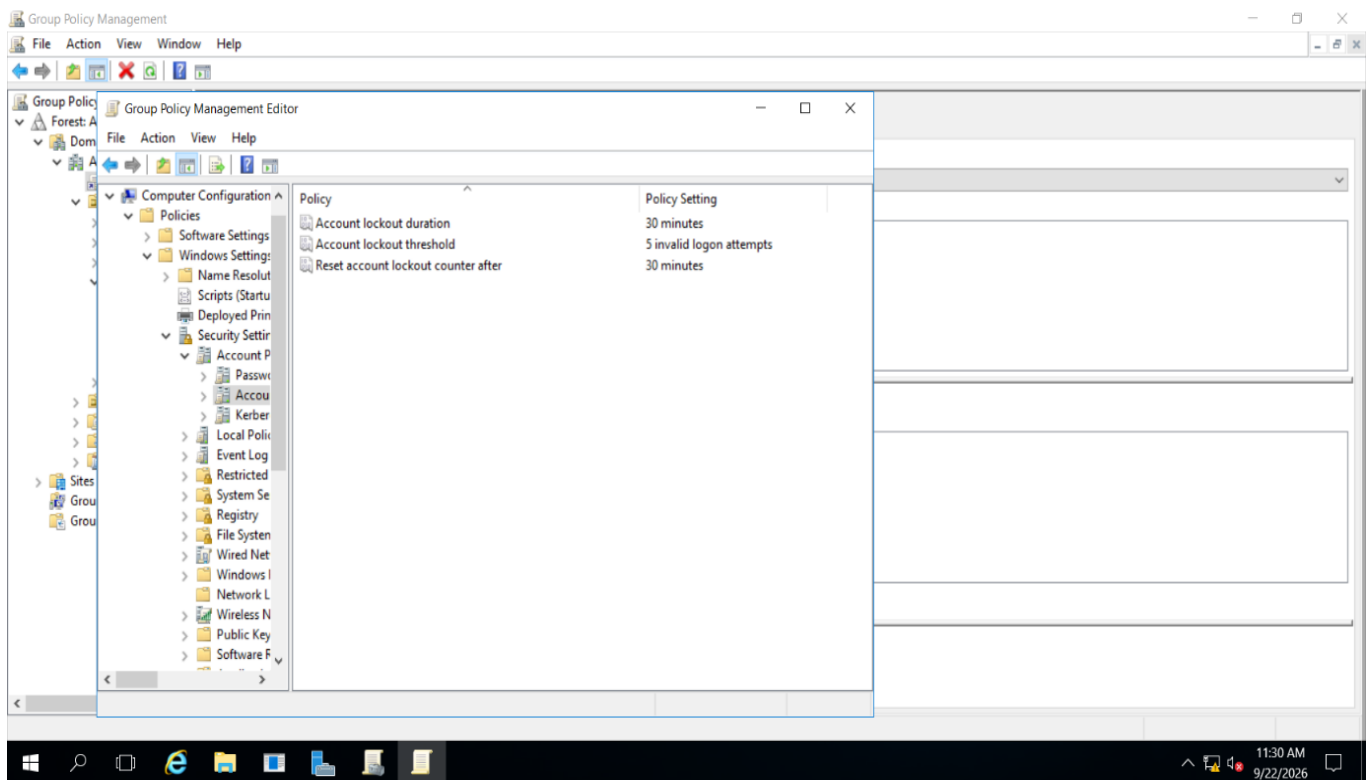
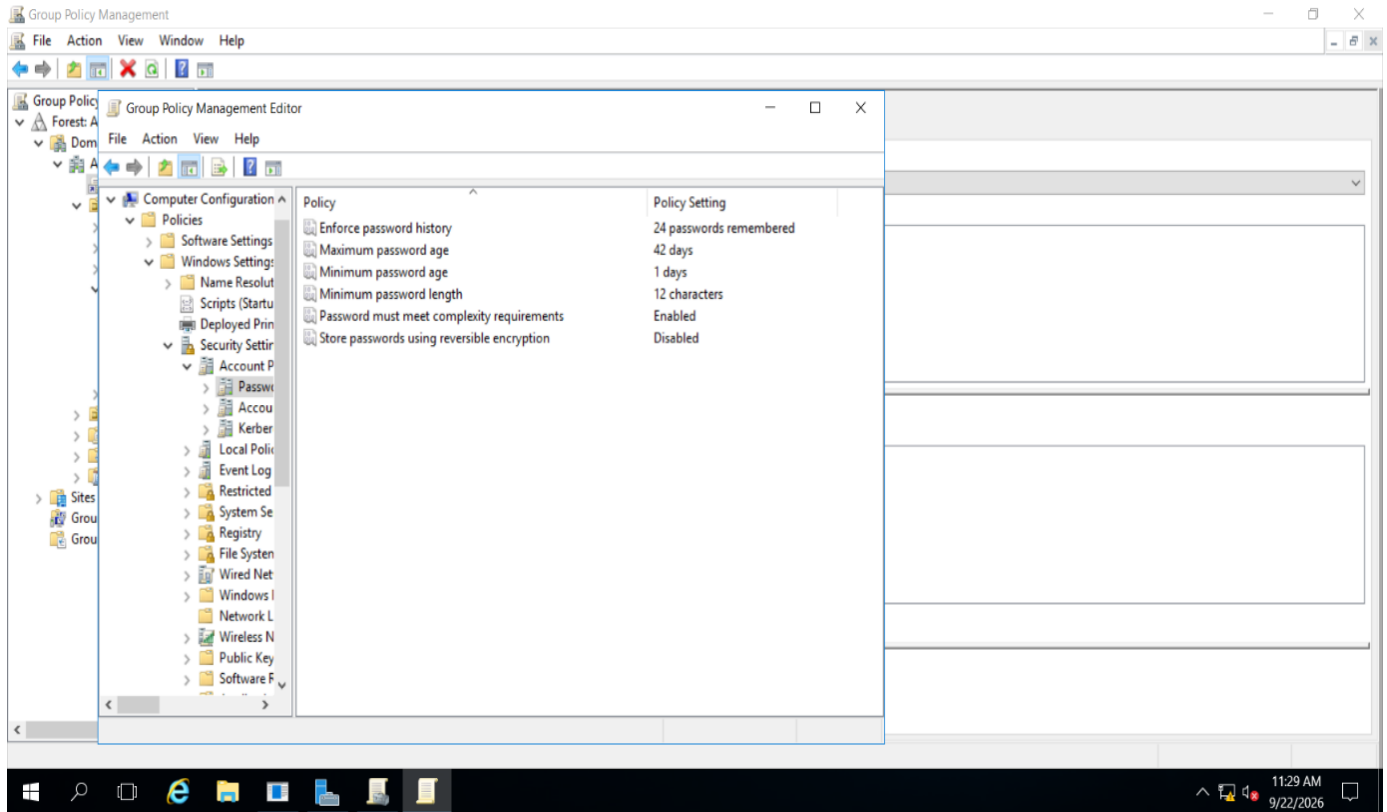


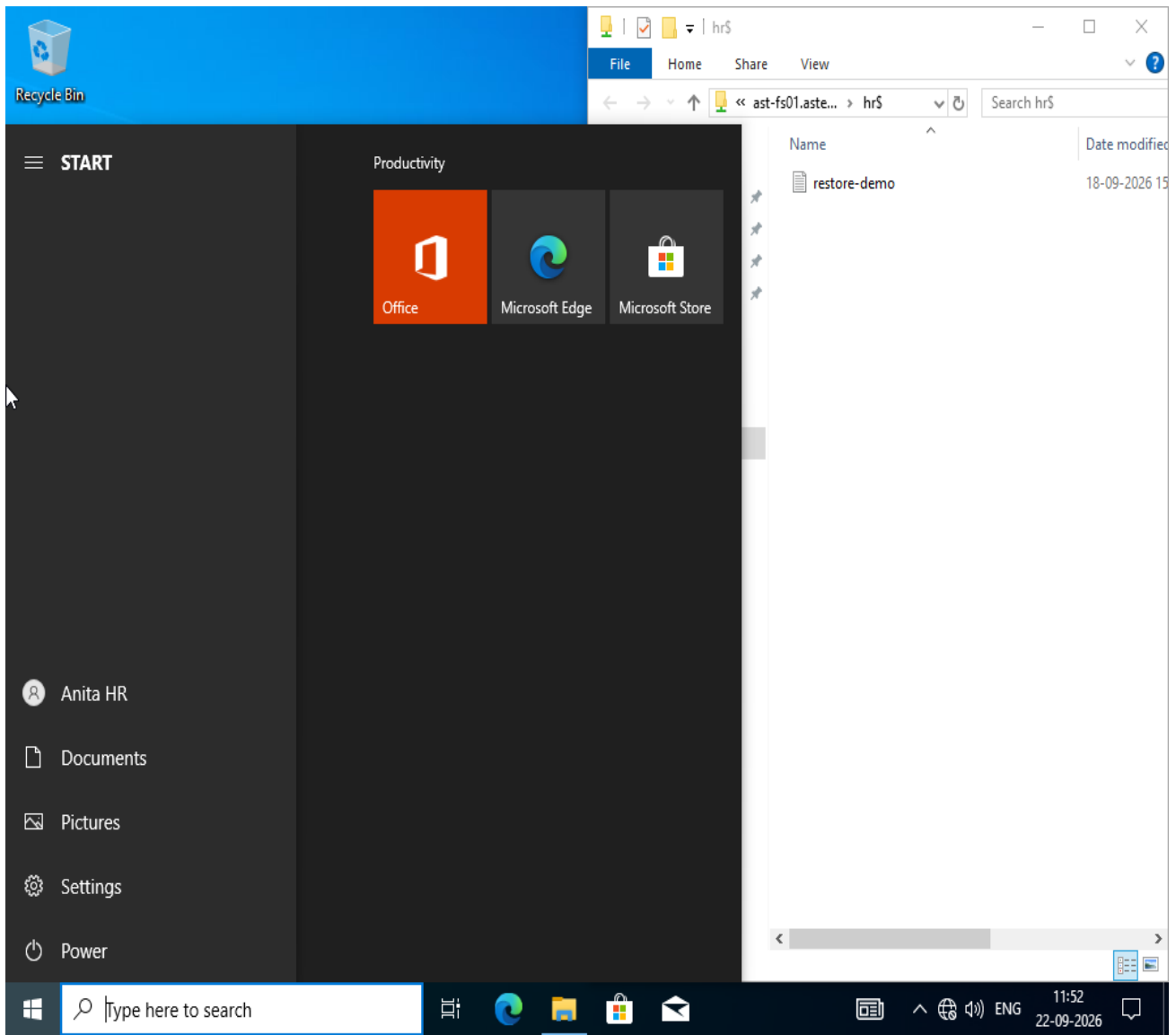


B14 | Install and configure DHCP



B15 | Set domain account policy and HR user policy





P2-01 | Build AST-FS02

The screenshot displays the Windows Server Manager interface for a local server named AST-FS02. The left-hand navigation pane includes links to the Dashboard, Local Server (which is currently selected), All Servers, and File and Storage Services. The main area is divided into two sections: PROPERTIES and EVENTS.

PROPERTIES For AST-FS02

Computer name		Last installed updates	
Computer name	AST-FS02	Last installed updates	Never
Domain	Aster.com	Windows Update	Download
		Last checked for updates	Never
Windows Firewall	Domain: On	Windows Defender	Real-Time
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC-08:00)
Ethernet0	192.168.50.21, IPv6 enabled	Product ID	Not active
Operating system version	Microsoft Windows Server 2016 Datacenter Evaluation	Processors	11th Gen I
Hardware information	VMware, Inc. VMware7,1	Installed memory (RAM)	2 GB
		Total disk space	39.45 GB

EVENTS
All events | 4 total

Server Name	ID	Severity	Source	Log	Date and Time
AST-FS02	1014	Warning	Microsoft-Windows-DNS Client Events	System	9/21/2026 11:10:08 PM
AST-FS02	8018	Warning	Microsoft-Windows-DNS Client Events	System	9/21/2026 11:10:03 PM
AST-FS02	6008	Error	EventLog	System	9/21/2026 11:09:54 PM

The Windows taskbar at the bottom shows the Start button, search icon, task view icon, and several application icons (Edge, File Explorer, etc.). The system clock indicates the time is 11:15 PM on 9/21/2026.

P2-02 | Join the second file server and install storage roles

System Properties

Computer Name/Domain Changes

You can change the name and the membership of this computer. Changes might affect access to network resources.

Computer name:
AST-FS02

Full computer name:
AST-FS02.Aster.com

More...

Member of

☒ Domain:
Aster.com

☐ Workgroup:

OK Cancel Apply

Server Manager

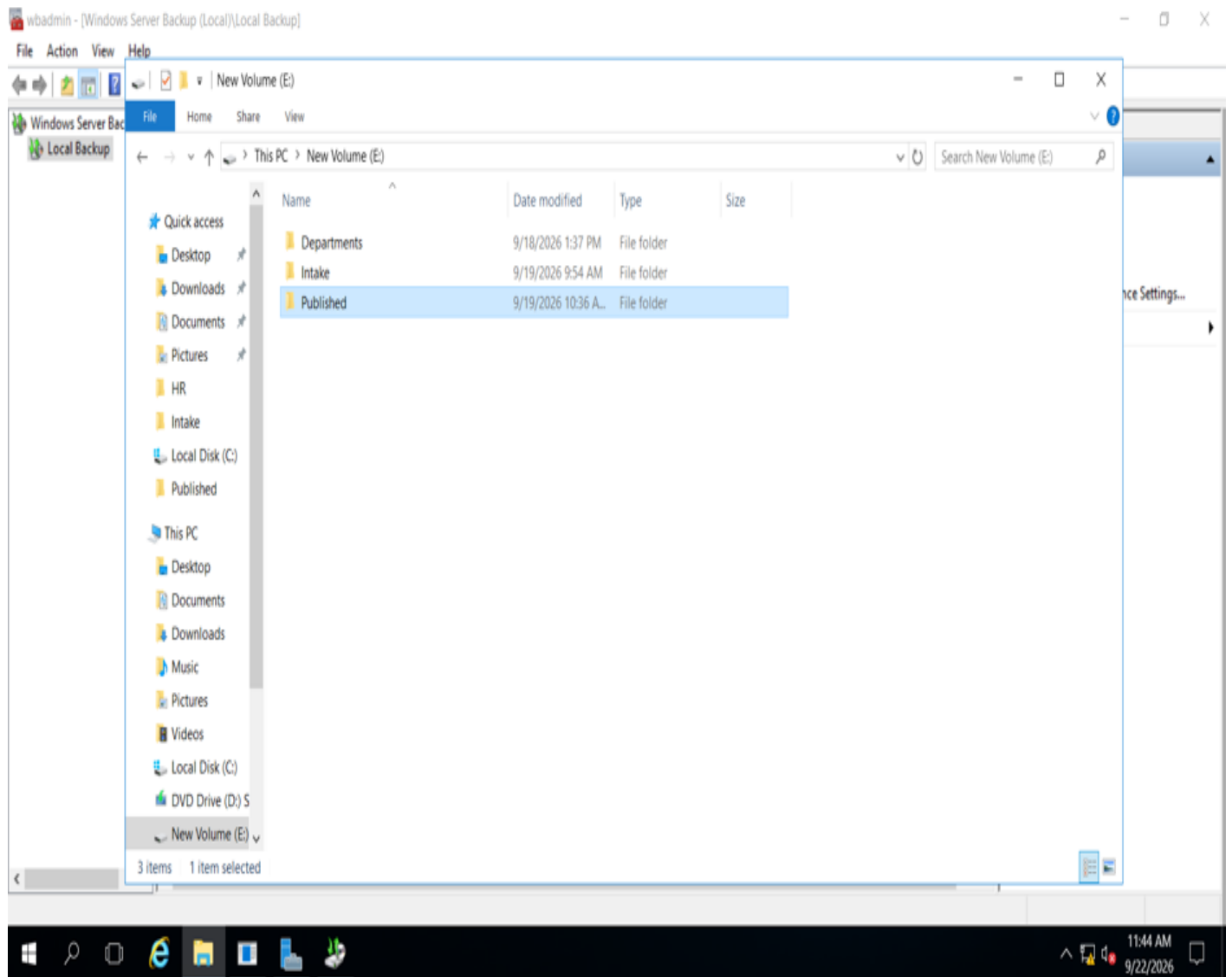
Manage Tools View Help

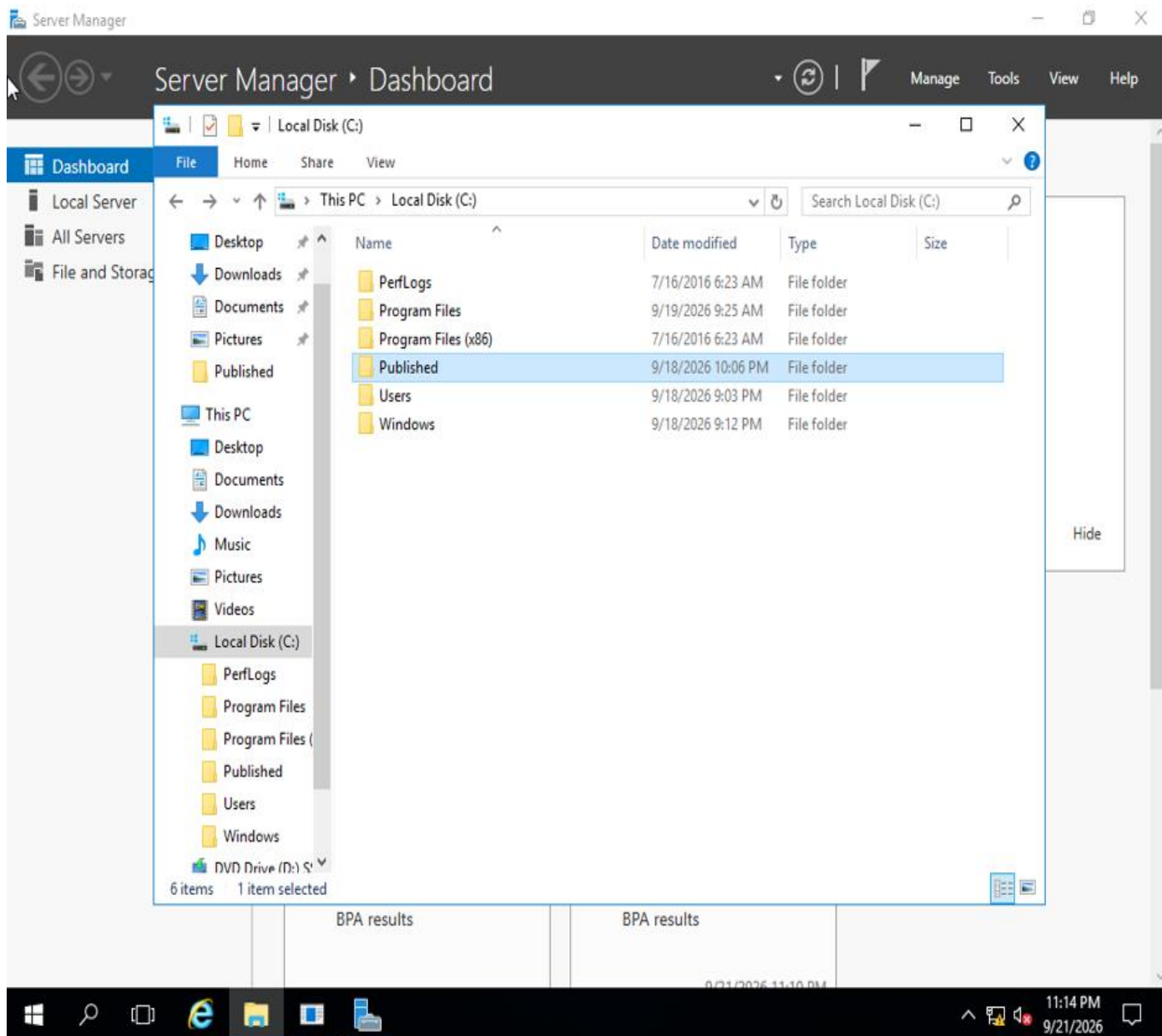
TASKS

Server Name	ID	Severity	Source	Log	Date and Time
AST-FS02	8198	Error	Microsoft-Windows-Security-SPP	Application	9/22/2026 1:40:43 AM
AST-FS02	1014	Error	Microsoft-Windows-Security-SPP	Application	9/22/2026 1:40:43 AM
AST-FS02	8200	Error	Microsoft-Windows-Security-SPP	Application	9/22/2026 1:40:43 AM

1:40 AM
9/22/2026

P2-03 | Create intake and published shares





P2-04 | Apply and prove an intake quota

File Server Resource Manager

File Action View Help

File Server Resource Manager (Local)

- Quota Management
 - Quotas
 - Quota Templates
- File Screening Management
- Storage Reports Management
- Classification Management
- File Management Tasks

Filter: Show all: 1 items

Quota Path	% Used	Limit	Quota Type	Source Template	Match Template	Description
Source Template: 100 MB Limit (1 item)						
E:\Intake	0%	100 MB	Hard	100 MB Limit	Yes	

Actions

- Quotas
 - Create Quota...
 - Refresh
 - View
 - Help

10:22 AM
9/23/2026

P2-05 | Block selected intake file extensions

File Server Resource Manager

File Action View Help

File Server Resource Manager (Local)

- Quota Management
 - Quotas
 - Quota Templates
- File Screening Management
 - File Screens**
 - File Screen Templates
 - File Groups
- Storage Reports Management
- Classification Management
- File Management Tasks

Filter: Show all: 1 items

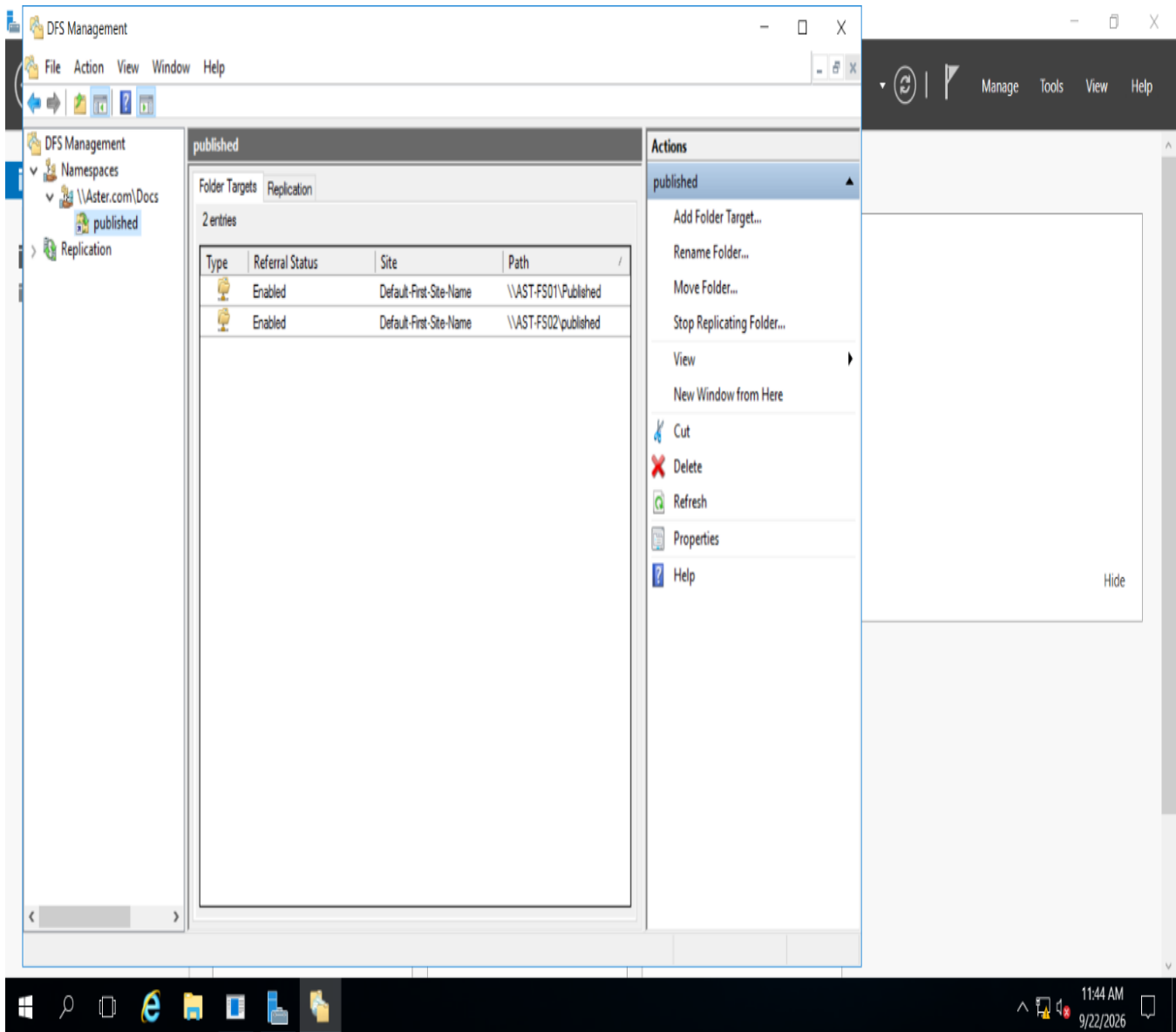
File Screen Path	Screening Type	File Groups	Source Template	Match Te...
Source Template: intake file screen (1 item)				
E:\Intake	Active	Block: Audio and Video Files, Executab...	intake file screen	Yes

Actions

- File Screens
 - Create File Screen...
 - Create File Screen Exception...
 - Refresh
 - View
 - Help

10:22 AM
9/23/2026

P2-06 | Create a namespace with two folder targets



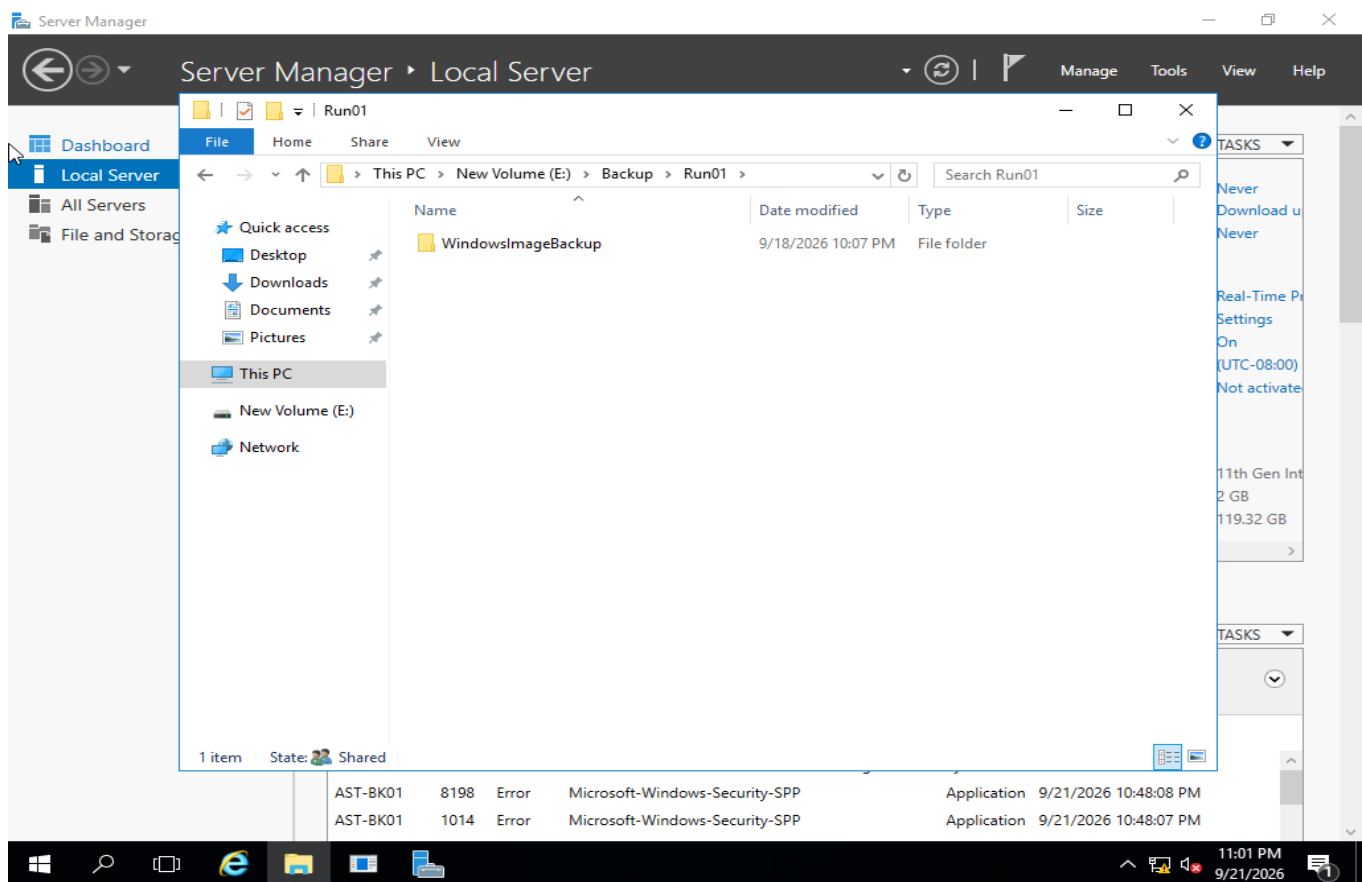
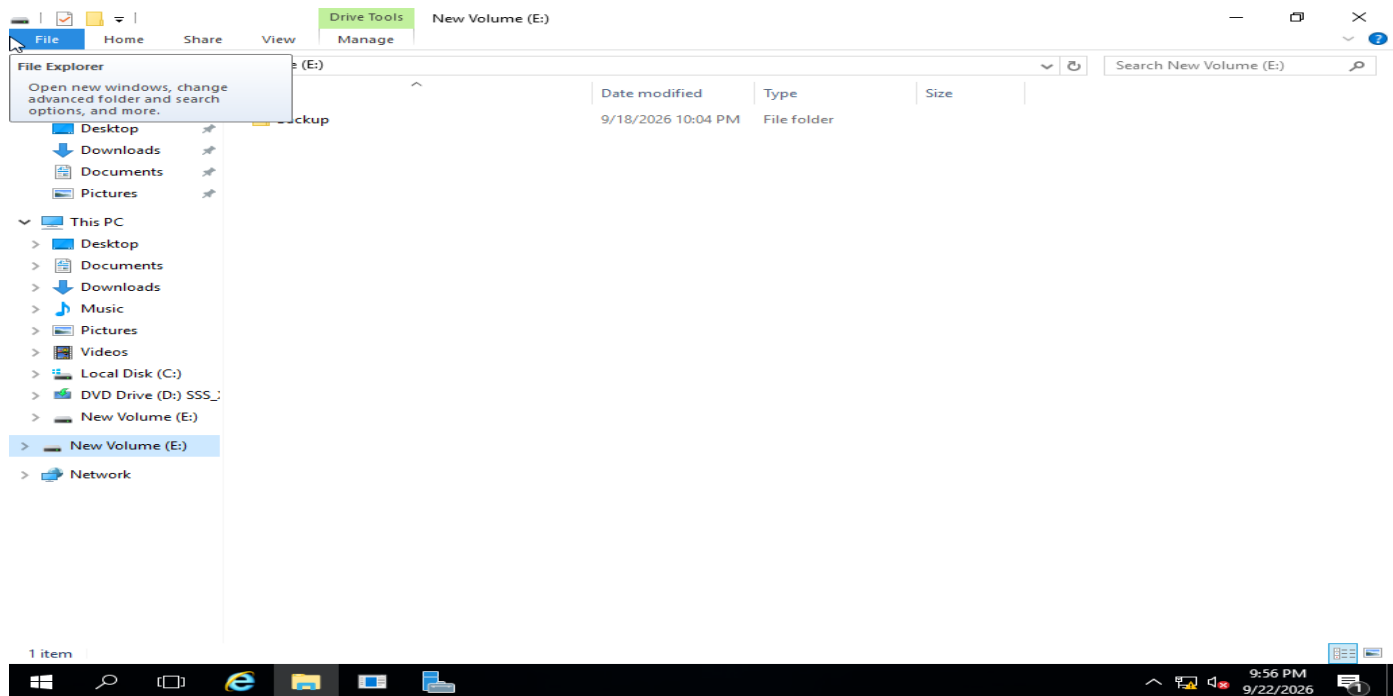
P2-07 | Configure replication and prove convergence

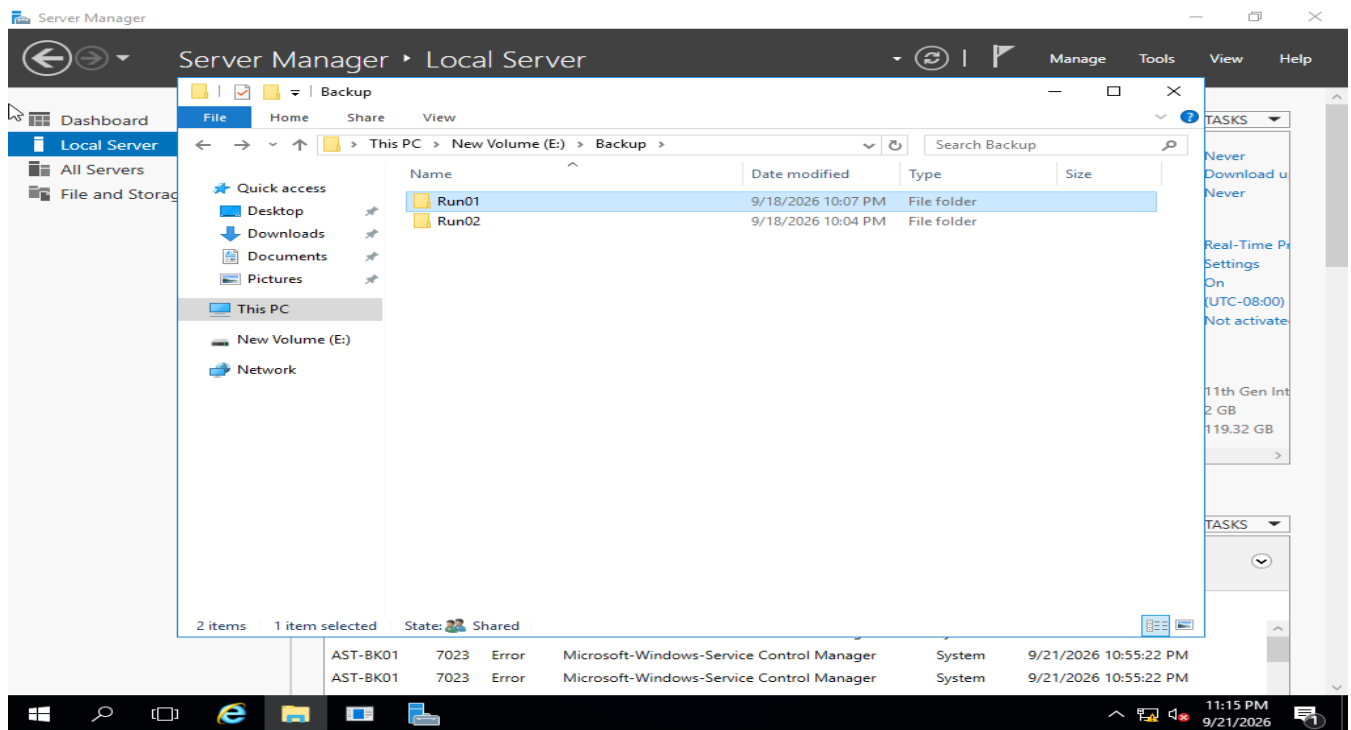
The screenshot shows the DFS Management console window. The left pane displays the hierarchy: DFS Management > Namespaces > Replication > aster.com\docs\published. The main pane shows the 'Replicated Folders' tab for 'aster.com\docs\published (Aster.com)'. It indicates '2 entries' and displays a table with the following data:

State	Local Path	Membership Status	Member	Replicated Folder	Staging Quota
Replicated Folder: published (2 items)					
	e:\Published	Enabled	AST-FS01	published	4.00 GB
	c:\Published	Enabled	AST-FS02	published	4.00 GB

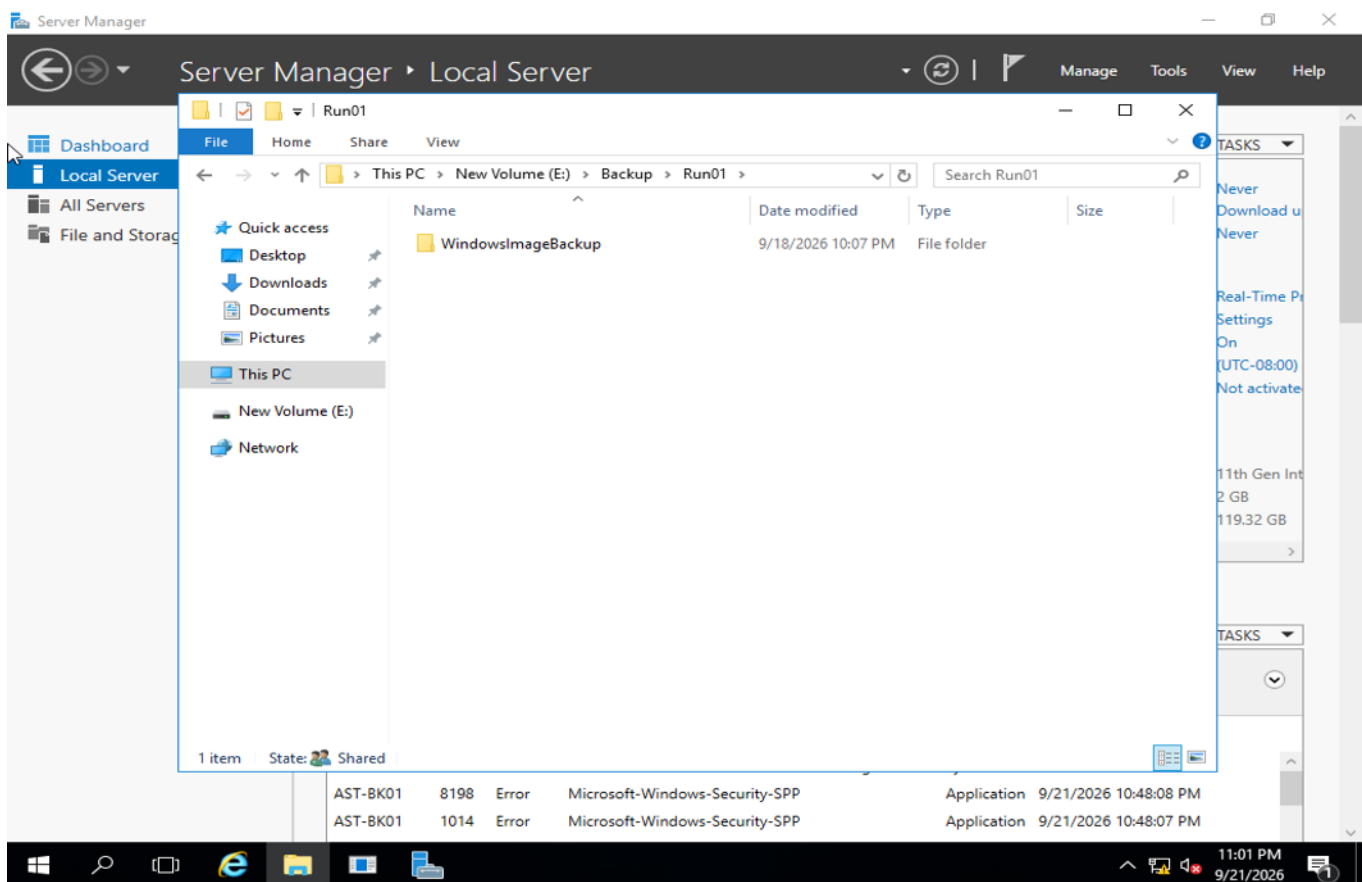
The right pane shows the 'Actions' menu for 'aster.com\docs\published', including options like 'New Member...', 'New Replicated Folders...', 'New Connection...', 'New Topology...', 'Create Diagnostic Report...', 'Verify Topology...', 'Delegate Management Permissions...', 'Edit Replication Group Schedule...', 'Remove Replication Group from Di...', 'View', 'New Window from Here', 'Delete', 'Refresh', 'Properties', and 'Help'.

P2-08 | Prepare a restricted backup destination





P2-09 | Create test data and the first backup



P2-10 | Delete and restore a file with integrity proof

wbadmin - [Windows Server Backup (Local)\Local Backup]

File Action View Help

Windows Server Backup (Local Backup)

Local Backup

You can perform a single backup or schedule a regular backup using this application.

A scheduled backup has not been configured for this computer. Use the Backup Schedule Wizard to set up a regular, automated backup to protect your data.

Messages (Activity from last week, double click on the message to see details)

Time	Message	Description
9/19/2026 10:37 AM	Backup	Successful
9/19/2026 10:31 AM	File recovery	Successful
9/19/2026 10:15 AM	Backup	Successful
9/18/2026 3:40 PM	File recovery	Successful
9/18/2026 3:15 PM	Backup	Successful

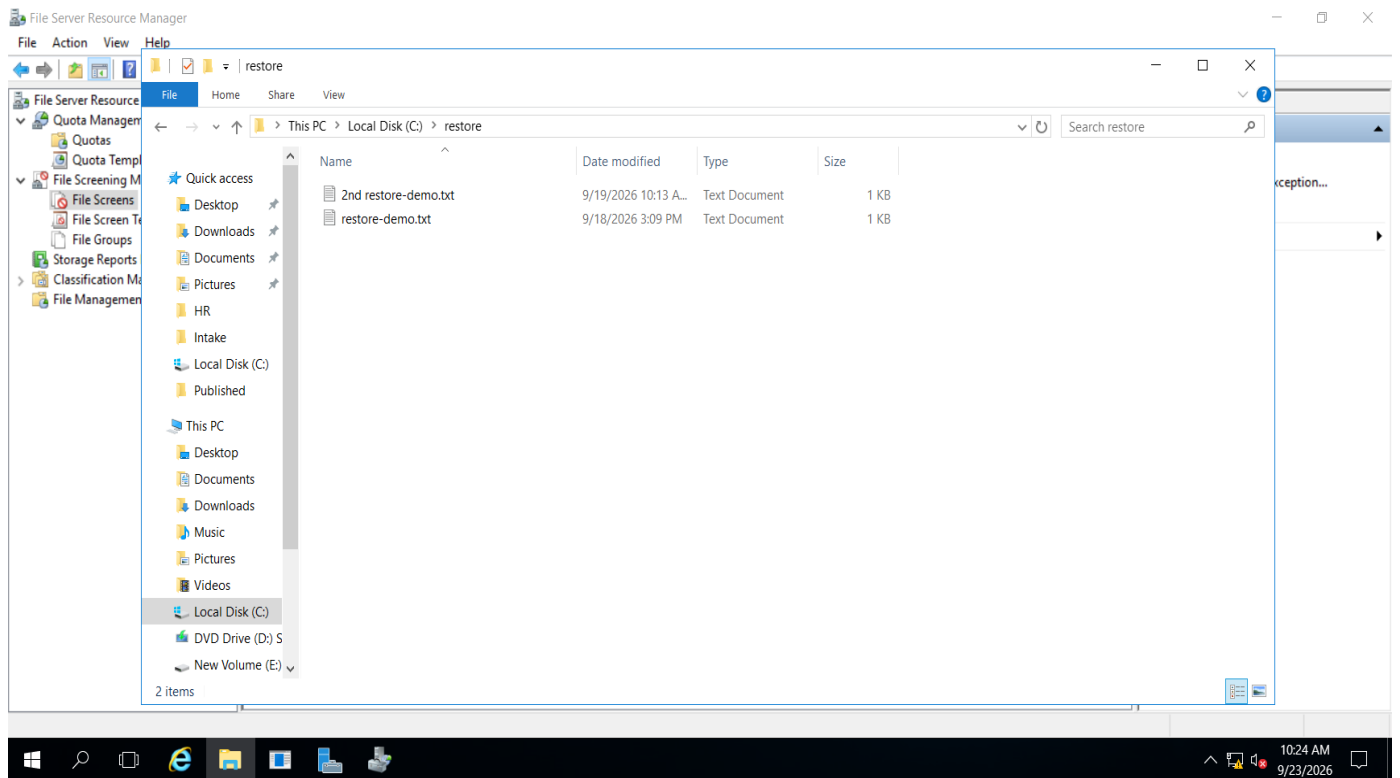
Status

Last Backup	Next Backup	All Backups
Status: Successful	Status: Not scheduled	Total backups: 2 copies
Time: 9/19/2026 10:37 AM	Time: -	Latest copy: 9/19/2026 10:37 AM
View details	View details	Oldest copy: 9/19/2026 10:15 AM
		View details

Actions

- Local Backup
- Backup Schedule...
- Backup Once...
- Recover...
- Configure Performance Settings...
- View
- Help

11:32 AM 9/22/2026



Project Summary

The project successfully demonstrated the implementation of a controlled document management environment using Windows Server and Active Directory.

Users were assigned access through security groups, while NTFS and share permissions were configured to control access to departmental resources.

Access validation was performed using different user accounts to confirm that authorized users could access their assigned resources and unauthorized users were restricted.

Project Type

Hands-on Training / Lab Project

Focus Areas:

Windows Server | Active Directory | File Server | NTFS Permissions | SMB | User & Group Management | Access Control | Troubleshooting